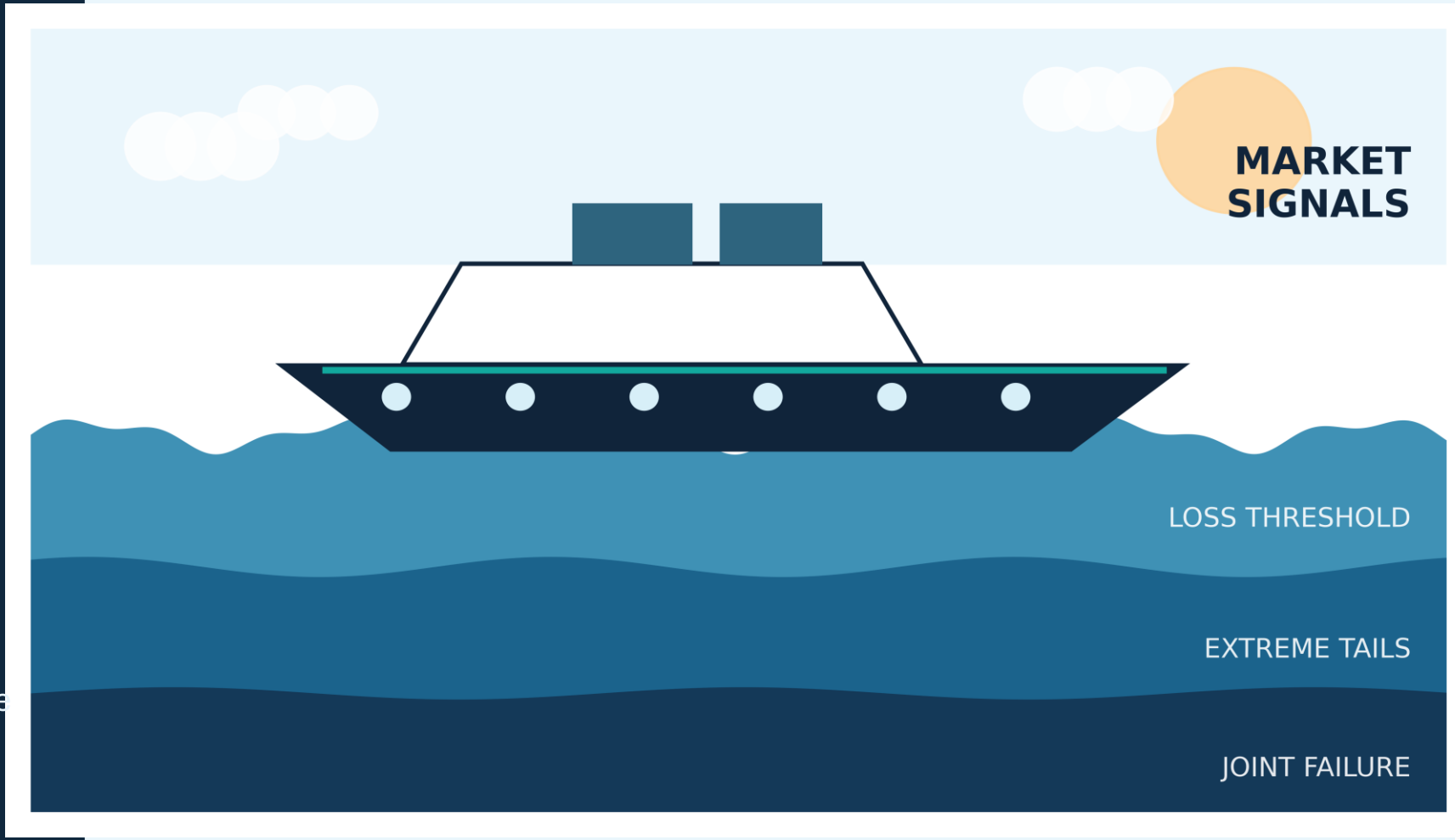
SEE RISK BEYOND CORRELATION

ADVANCED QUANTITATIVE RISK MANAGEMENT

Ship analogy • VaR • ES • EVT • Wavelets • Copulas • Institutional case



All numerical illustrations are synthetic unless indicated

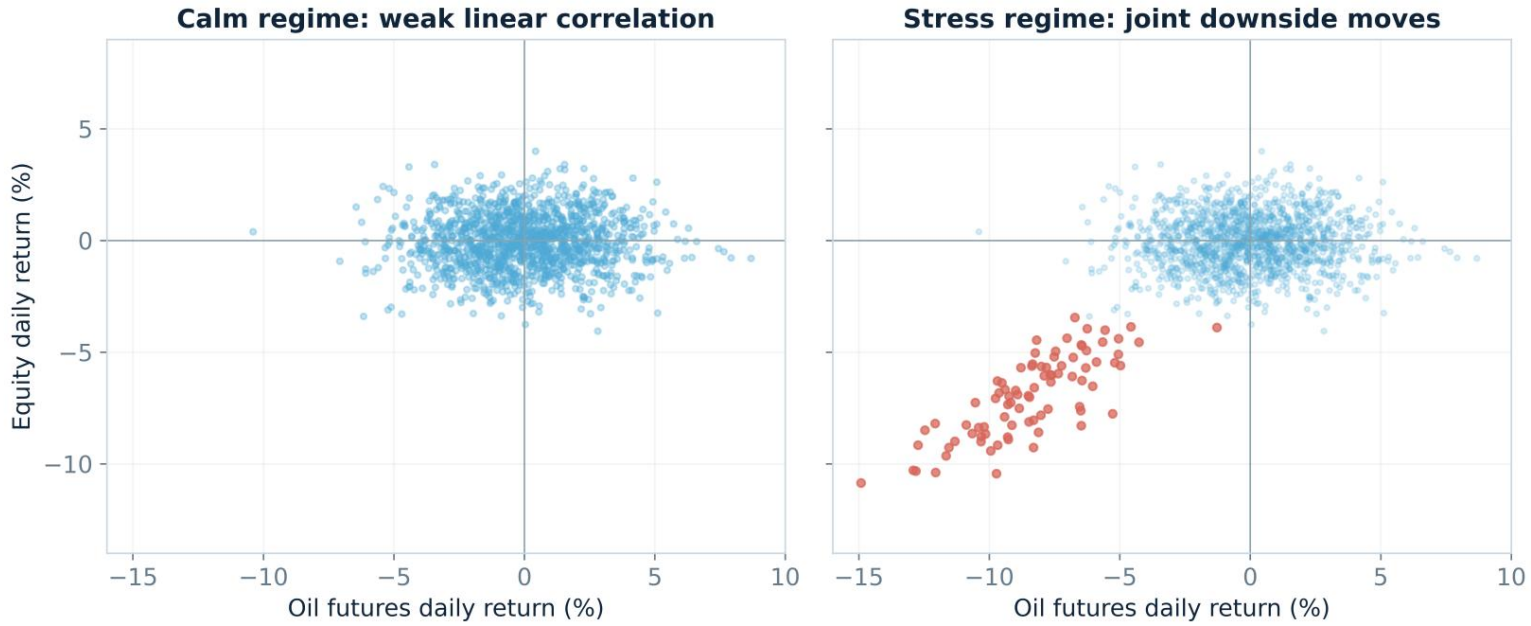
From financial intuition to risk decisions

Five linked questions guide the whole presentation.

01	What is economically at risk?	Oil futures, equity, notional and margin
02	What signal matters?	Wavelets and investment horizons
03	How large can losses become?	Value-at-Risk and Expected Shortfall
04	What happens in the extremes?	EVT and copula joint crashes
05	What should managers change?	Limits, capital and liquidity buffers

Low correlation does not imply safety

Diversification can weaken precisely when liquidity and capital matter most.



Normal conditions

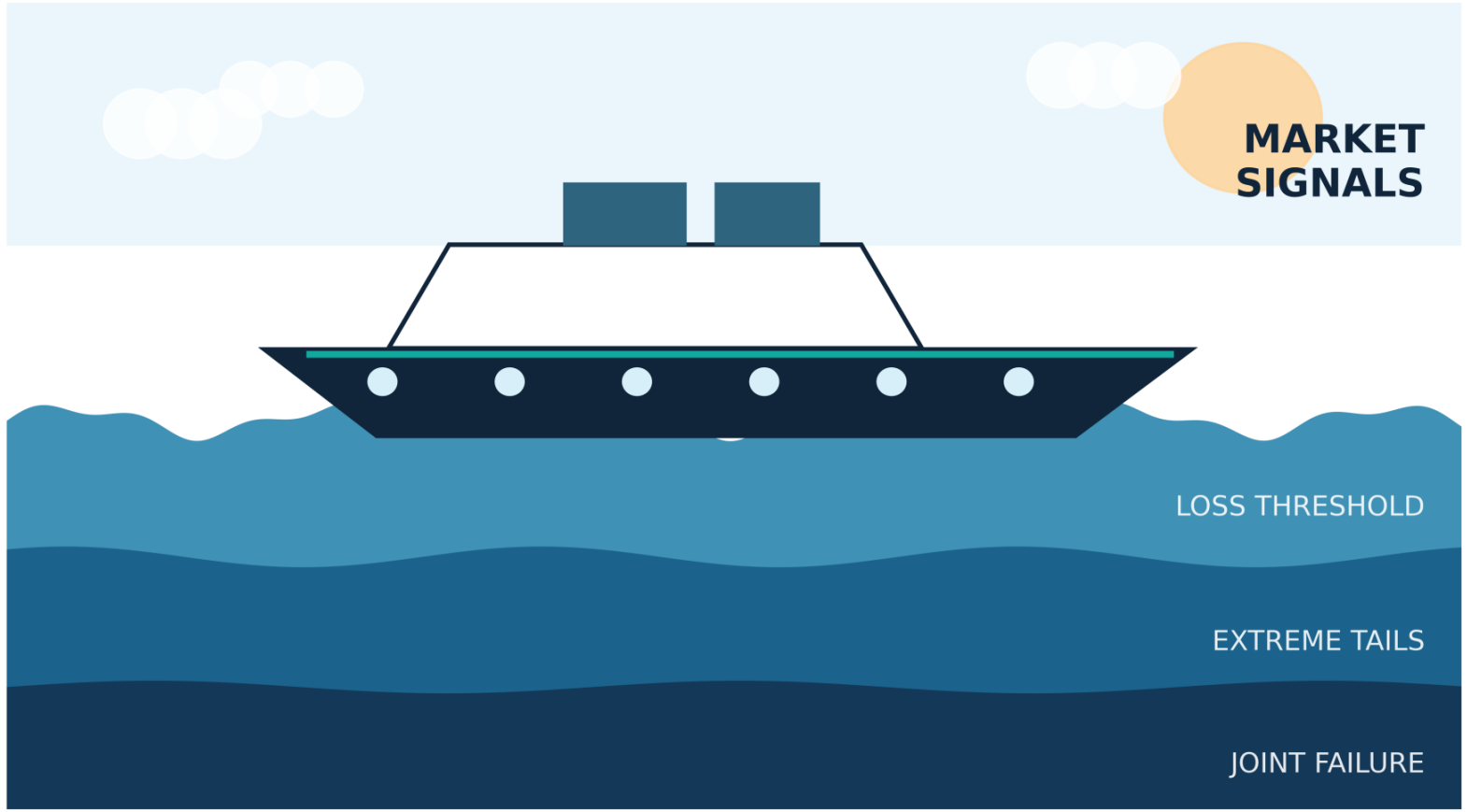
Returns can move almost independently. Pearson correlation sees the average linear pattern.

Stress conditions

Both assets may fall together, causing simultaneous losses and urgent margin calls.

A ship crossing a volatile financial ocean

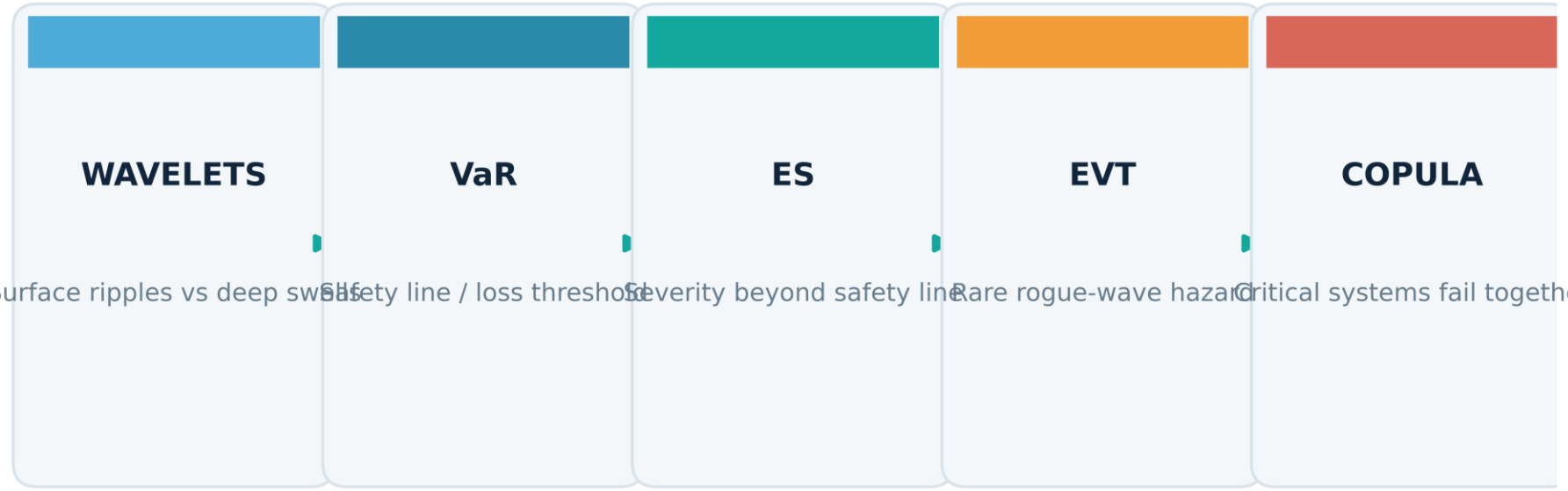
A mental model — not a literal mathematical equivalence.



- Waves = prices and returns changing through time
- Waterline = loss threshold measured by VaR
- Flooding = severity of losses captured by ES
- Rogue wave = rare event examined with EVT

Five risk tools, five different questions

Complementary tools: none should be treated as a complete risk model by itself.



Portfolio interpretation: signal → threshold → tail severity → extreme events → joint failures

Wavelets · VaR · ES · EVT · Copulas

What exactly is the fund holding?

Oil futures and equity holdings generate different cash flows and liquidity needs.

Crude oil futures (long)

Exchange-traded obligation: gain if the futures price rises; lose if it falls. Post initial margin and settle daily variation. A futures position is an exposure, not the full cash purchase of oil.

S&P 500 equity exposure (long)

A long equity holding benefits from rising share/index values and loses value when markets decline. Ownership or fund exposure is financed differently from futures margin.

Finance principle: measure economic exposure and liquidity requirements separately.

$10m futures notional is not $10m cash invested; margin is a performance bond.

The futures price drives the daily futures P&L

For a long position, use changes in the traded futures contract — not spot returns.

$$\Delta \Pi_{\text{fut}} = Q\,(F_t - F_{t-1}) \approx N\,r_F$$

Q = contract quantity

Contract multiplier × number of contracts, in underlying units.

F = futures price

Use matched contract maturity and daily settlement prices.

N = notional exposure

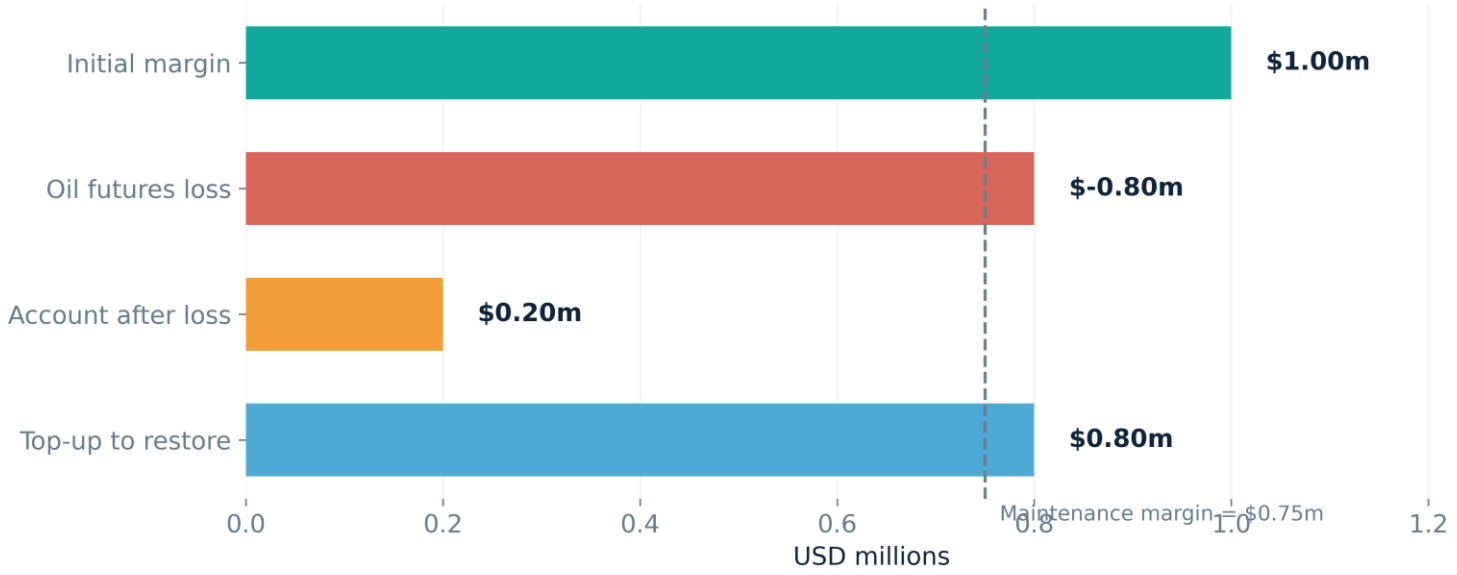
Approximate percentage price move × notional for small changes.

Important: rolling contracts, basis and term structure can make futures returns differ from spot returns.

Daily mark-to-market can create immediate cash calls before the investment thesis plays out.

An $800,000 futures loss can trigger a cash call

Illustrative long oil futures: $10m notional; initial margin $1m; maintenance $0.75m.



Daily settlement

An 8% oil futures decline creates a $0.80m mark-to-market loss on $10m notional.

Margin call

Margin equity drops to $0.20m, below $0.75m maintenance; a $0.80m top-up restores initial margin.

Two long positions, one liquidity shock

A combined decline can affect both wealth and the ability to maintain futures positions.

Oil futures

$10.00m notional × (−8%) = −$0.80m P&L. Cash is required through variation margin.

Equity exposure

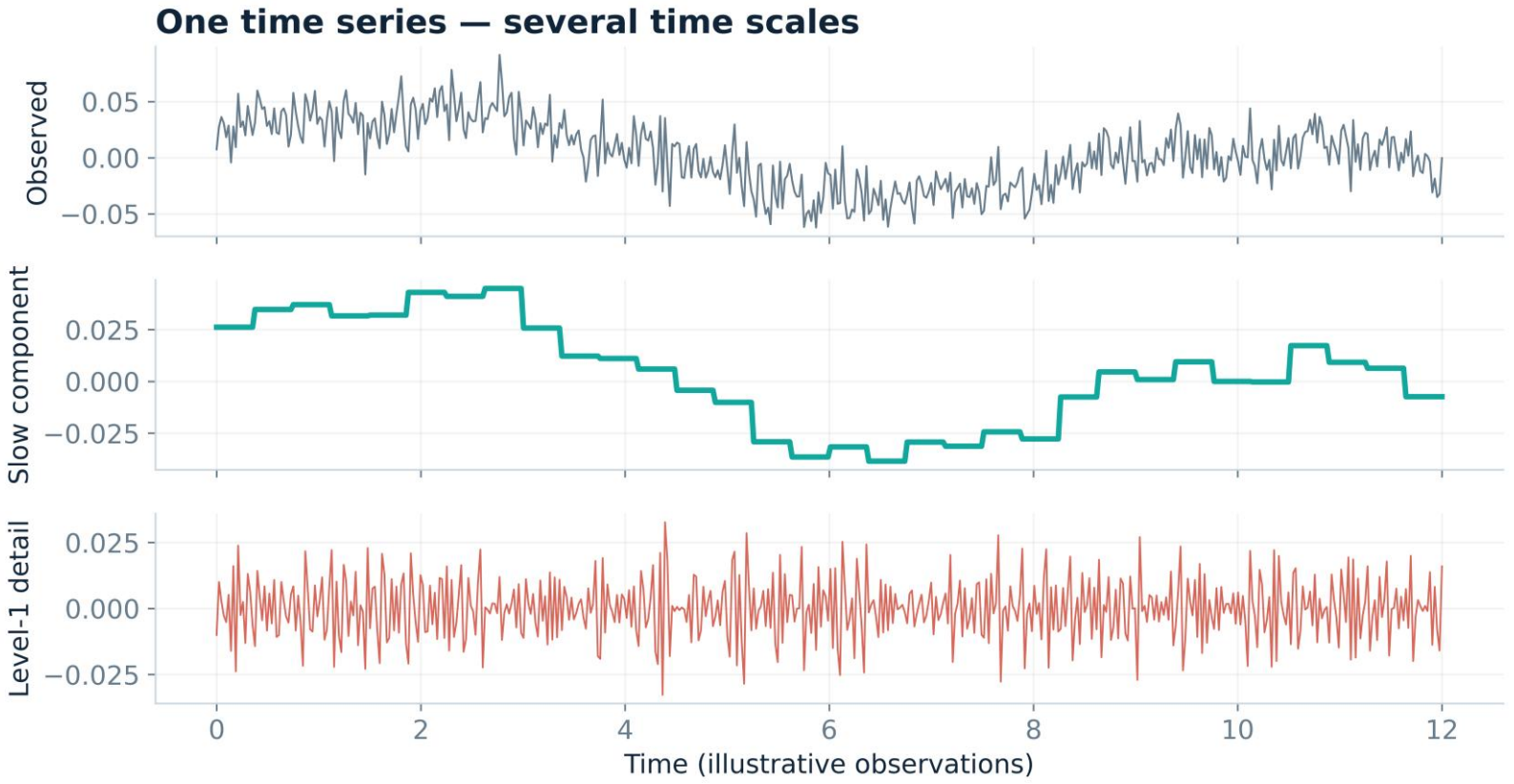
$10.00m market value × (−7%) = −$0.70m change in market value.

Combined portfolio P&L: −$1.50 million

- The $0.80m margin payment is cash settlement of the futures loss — not an extra loss.

A price signal contains more than one time scale

Separate short-run fluctuations from slower changes before interpreting risk.



Detail coefficients

Capture higher-frequency movements. They are not always “noise”; intraday jumps may be essential risk signals.

Approximation coefficients

Capture slower components, whose financial meaning depends on the sampling interval.

Decompose, reconstruct, then test

A DWT is a mathematical change of representation, not an automatic prediction of crises.

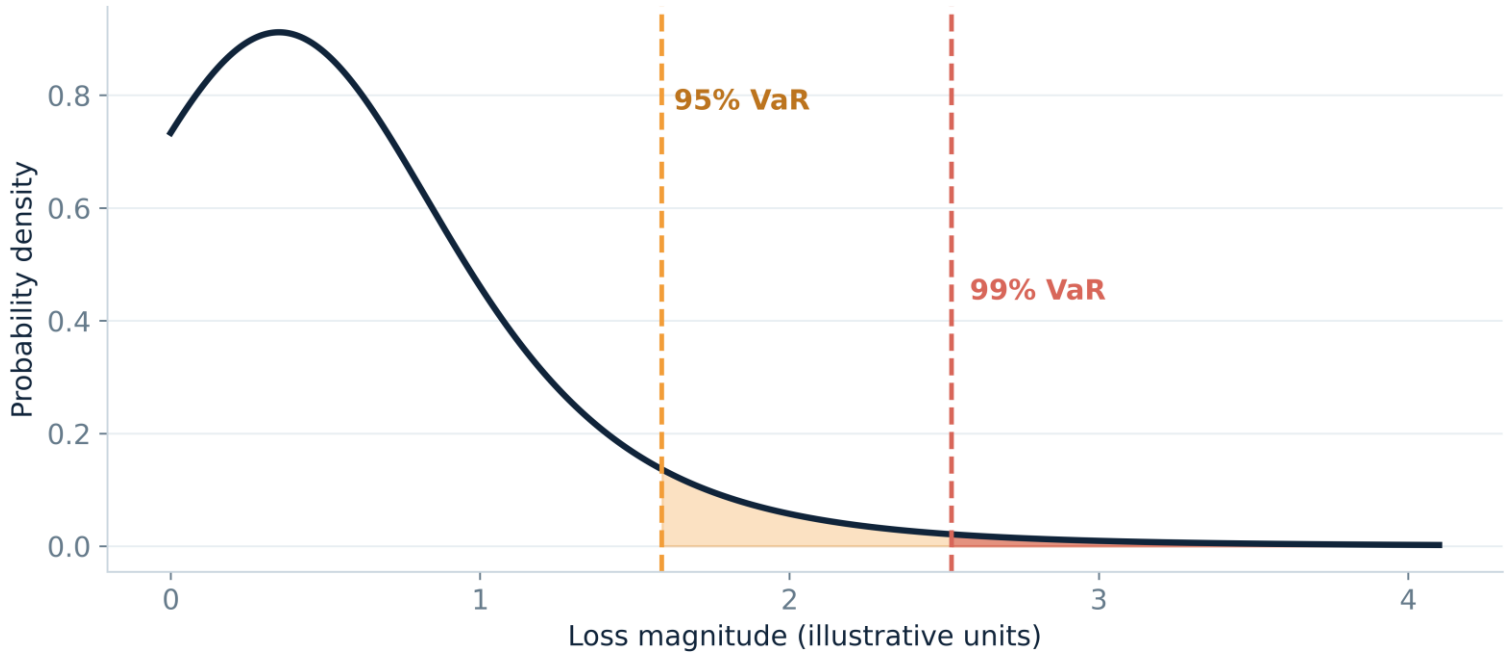
$$x(t) = \sum_{k} c_{J,k}\ \phi_{J,k}(t) + \sum_{j=1}^{J} \sum_{k} d_{j,k}\ \psi_{j,k}(t)$$

$$\psi_{j,k}(t) = 2^{j/2}\psi\left(2^{j}t - k\right)$$

- c(J,k): broad approximation / slow component
- d(j,k): detail at resolution j; choose basis and level from the data frequency
- Always verify that filtering has not erased economically important tail jumps.

VaR describes a loss percentile — not a worst loss

At 95% confidence, the loss threshold is exceeded on about 5% of days under the model.



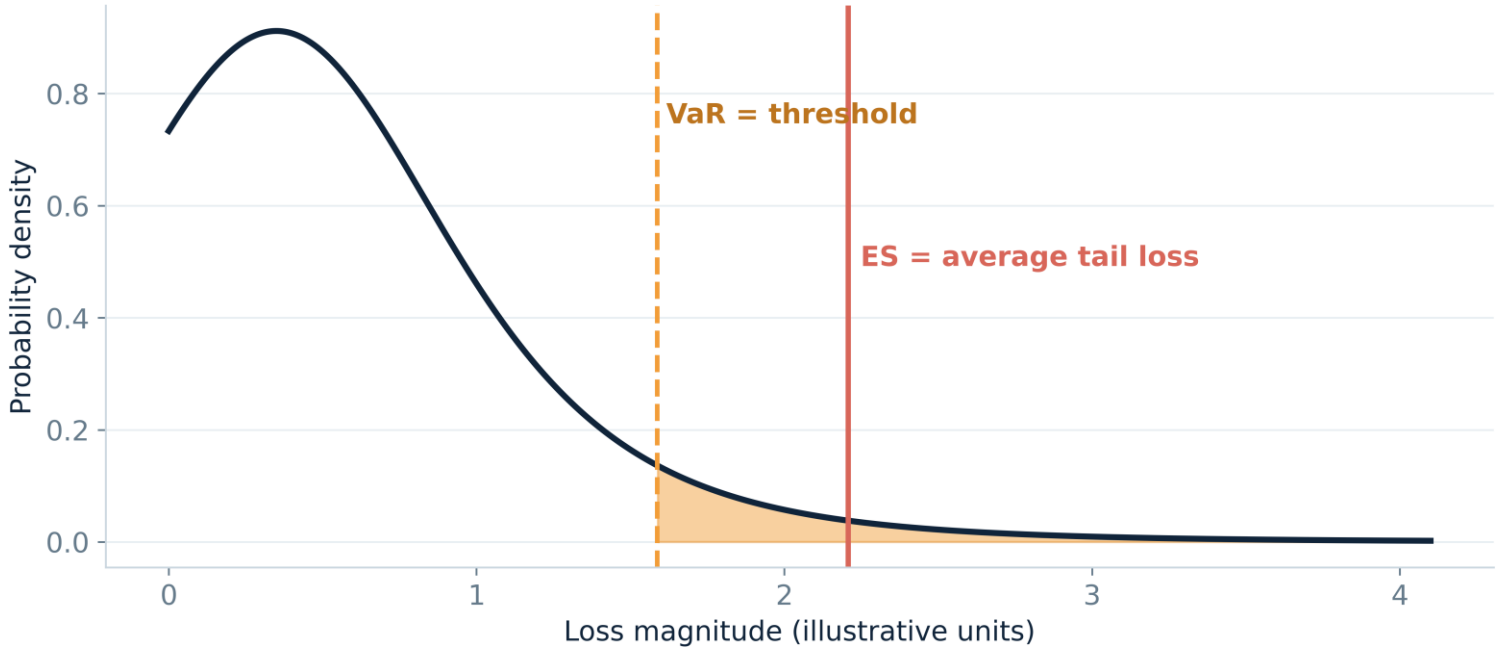
Confidence ≠ guarantee

VaR gives no direct information about average loss beyond its threshold.

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell \in \mathbb{R} : F_L(\ell) \geq \alpha\}$$

ES measures the average loss inside the worst tail

A coherent risk measure; it captures tail severity that VaR alone leaves out.



95% ES

Average loss among the worst 5% of outcomes for a continuous loss distribution.

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u$$

In general distributions, use the quantile-integral definition; simple conditional expectation needs care at point masses.

Two questions — threshold and consequence

Both depend on the time horizon, distribution, data and current exposures.

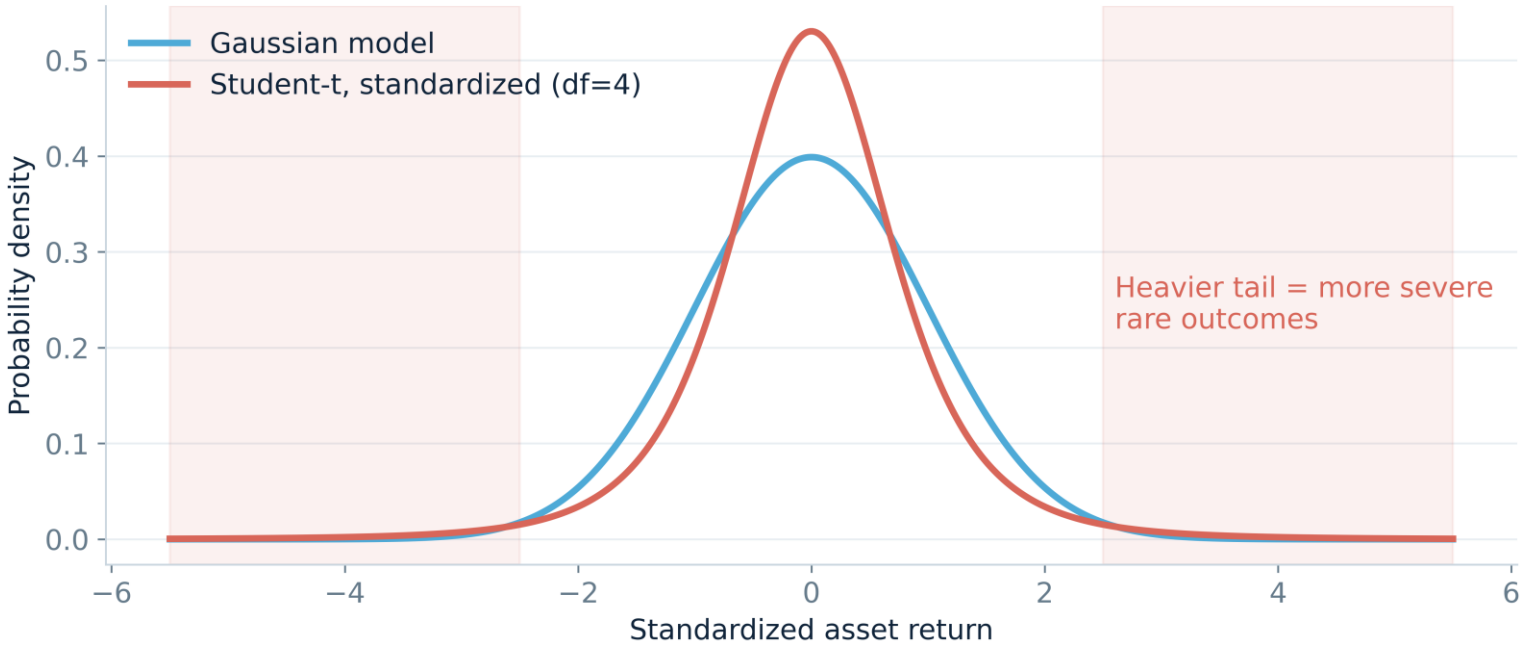
DIMENSION	VaR	ES
Question	Where is the loss cutoff?	How bad are losses beyond it?
Statistic	A quantile of losses	Average of the worst tail
Diversification	Not always subadditive	Coherent (subadditive)
Management use	Limits and monitoring	Capital and tail-loss budgeting

Neither risk metric predicts exactly when a market crisis will occur.

Backtesting and stress scenarios remain essential.

Normal models may understate extreme outcomes

Real return distributions can be skewed, clustered and heavy-tailed.



Fat-tail intuition

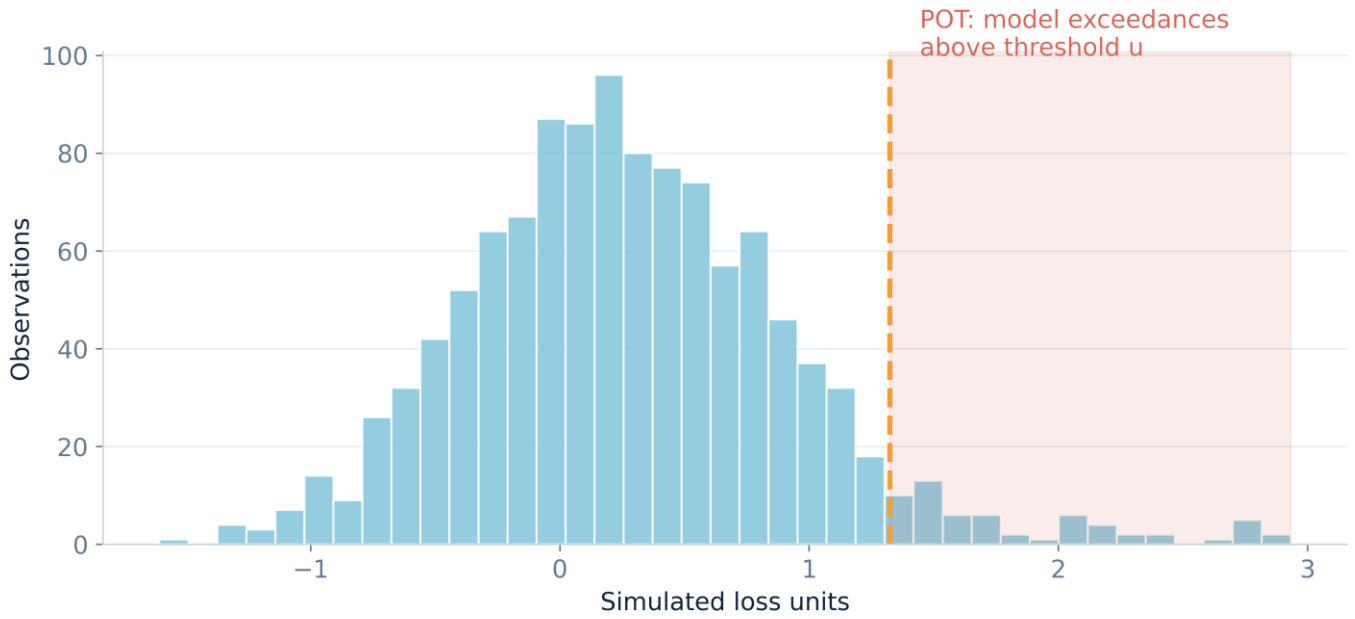
A heavy tail means extreme standardized outcomes are more probable than under a Gaussian benchmark.

EVT objective

Model the behavior of rare extremes rather than fitting one distribution to every ordinary return.

Model losses beyond a sufficiently high threshold

Select a threshold carefully; very few exceedances cause high estimation uncertainty.



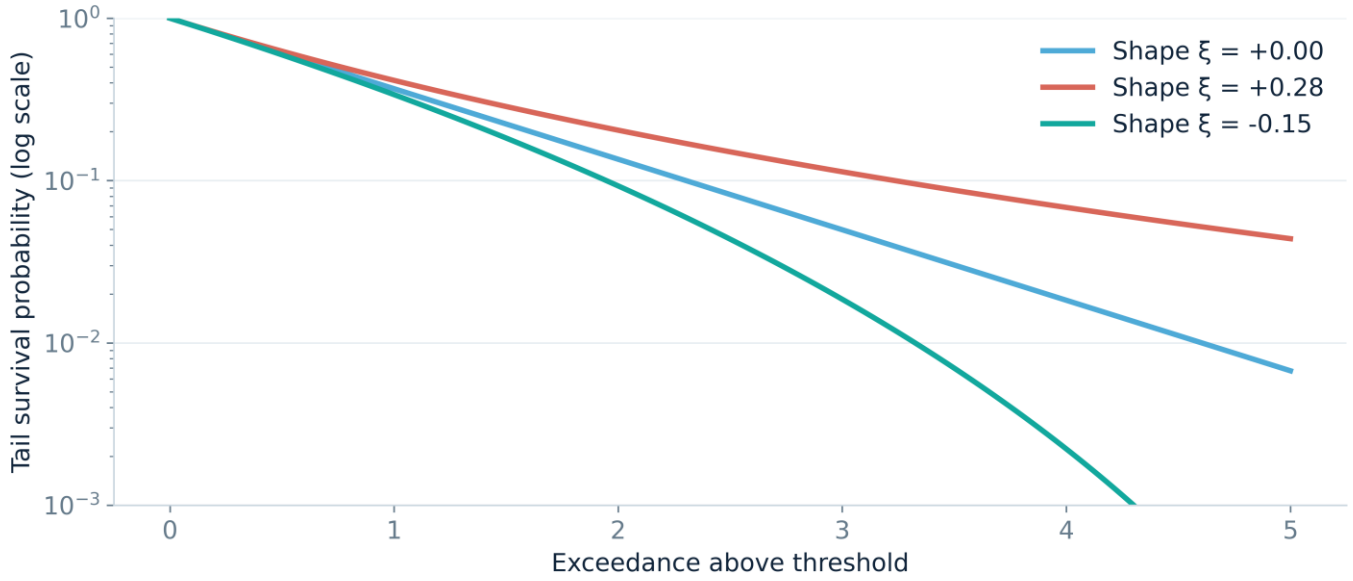
POT logic

Define exceedances y = L − u, only for L > u. Fit the GPD to those positive exceedances.

$$F_u(y) = \Pr(L - u \le y \mid L > u) = \frac{F_L(u + y) - F_L(u)}{1 - F_L(u)}$$

The shape parameter changes tail behavior

β is scale; ξ is shape. The support must satisfy 1 + ξy/β > 0.



$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi} \qquad (\xi \neq 0)$$

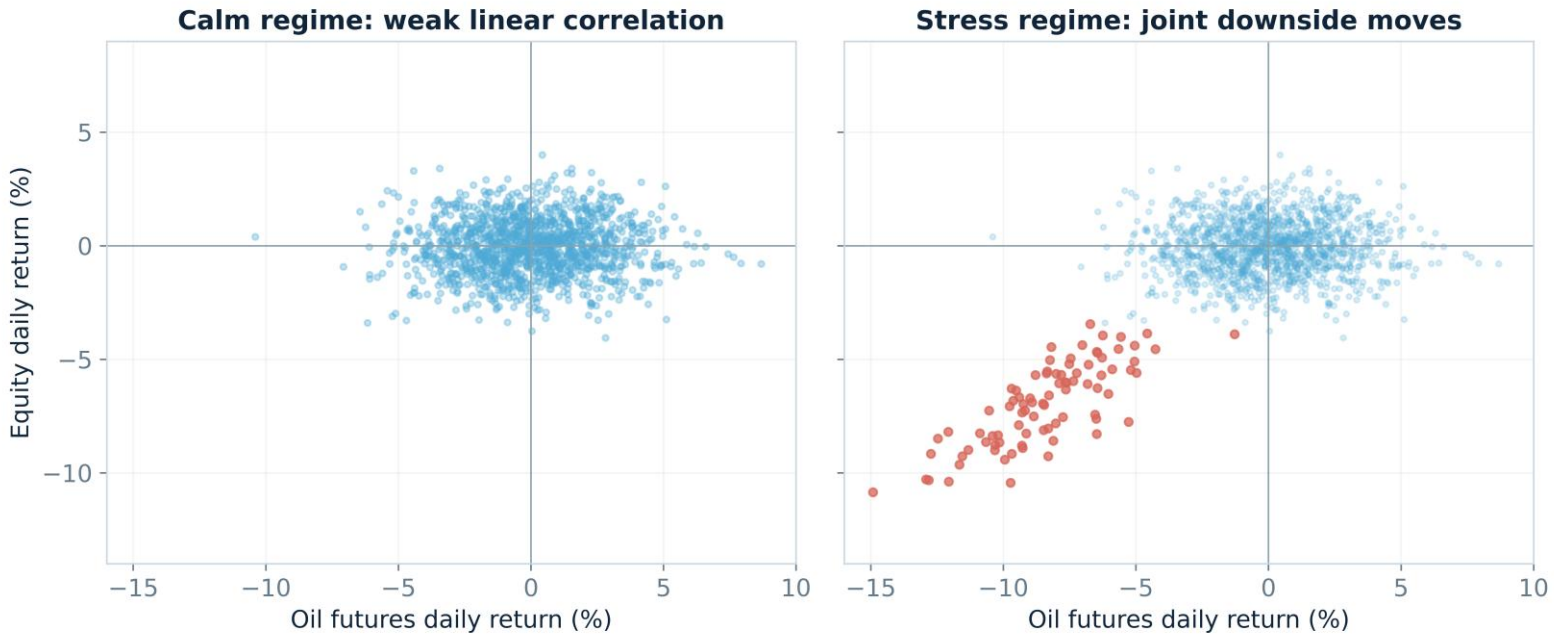
$$G_{0,\beta}(y) = 1 - \exp\left(-\frac{y}{\beta}\right)$$

Interpretation of ξ = +0.28

In a fitted illustrative model, positive ξ indicates a heavy, power-law tail. It does NOT establish a specific crash probability by itself.

Pearson correlation is an average — not a tail contract

Near-zero calm-market correlation does not rule out large common downside moves.

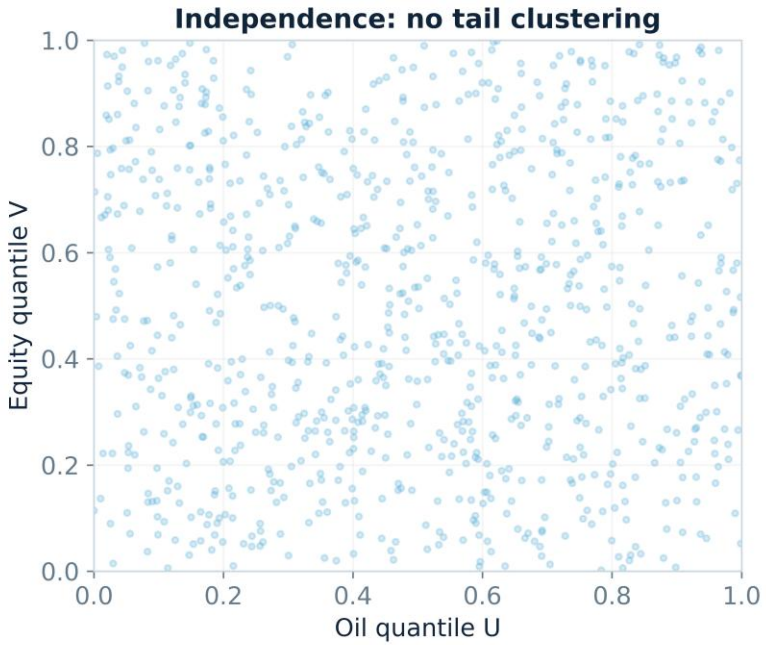


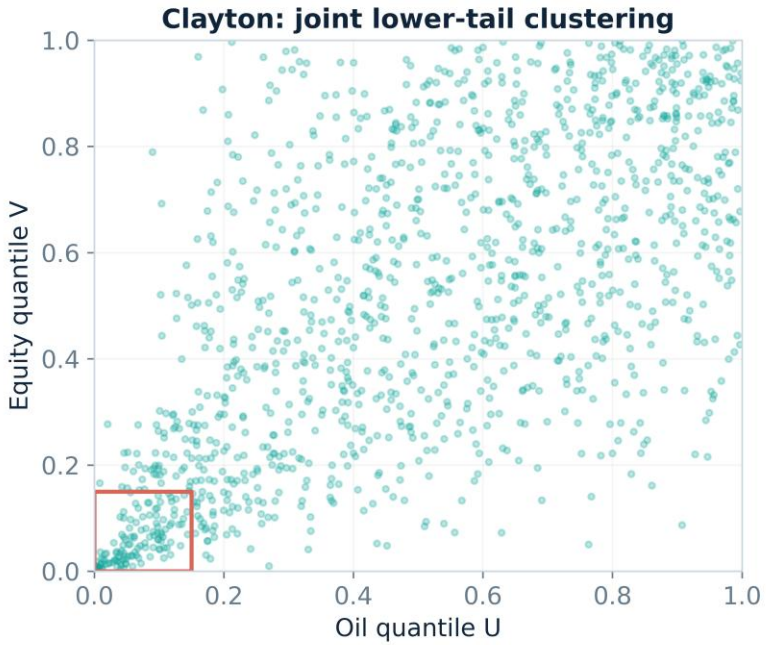
$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \, \sigma_Y}$$

- Correlation captures linear co-movement.
- Dependence can change across regimes.

Dependence is not the same as correlation

A copula links marginal distributions into a joint distribution.





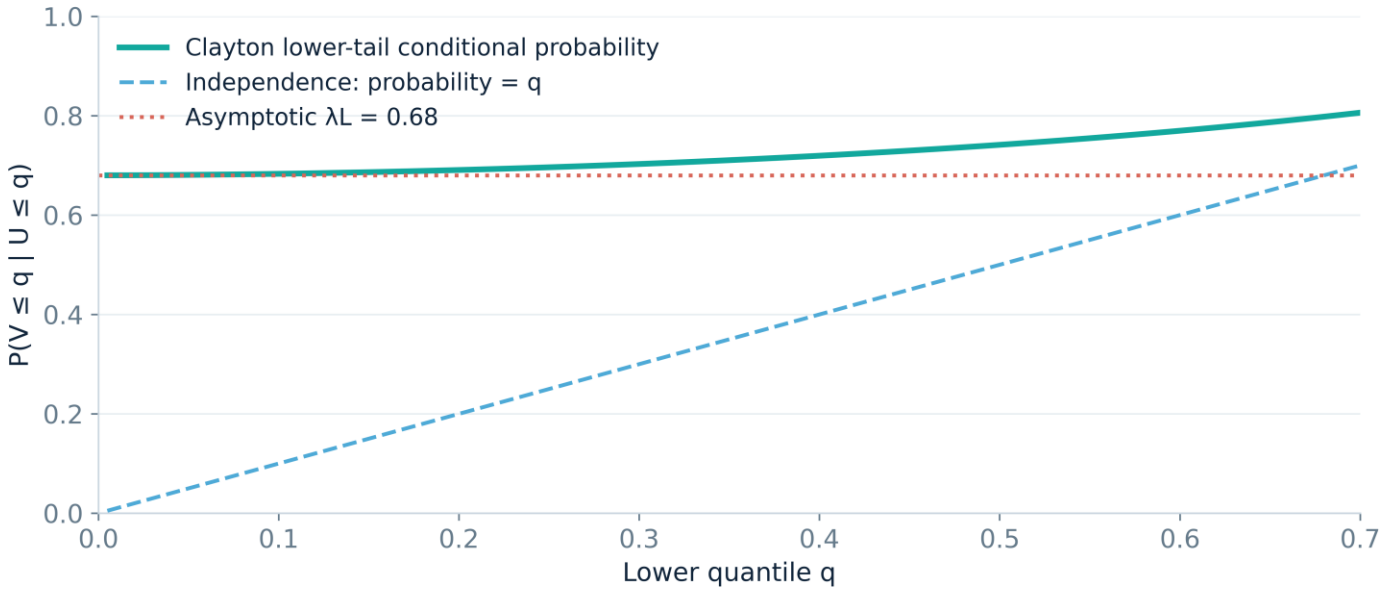
Sklar's theorem

For continuous marginals, the copula representation is unique. Tail dependence concerns joint extreme quantiles.

$$F_{X,Y}(x,y) = C\big(F_X(x), F_Y(y)\big)$$

Joint downside dependence can persist in rare events

Clayton is an asymmetric dependence model: lower-tail dependence and zero upper-tail dependence.



$$C_\theta(u,v) = \left(u^{-\theta} + v^{-\theta} - 1\right)^{-1/\theta}$$

$$\lambda_L = \lim_{q \downarrow 0} \frac{C(q,q)}{q} = 2^{-1/\theta}$$

- Illustration: λL = 0.68 implies θ ≈ 1.80.
- 68% is an asymptotic conditional tail measure, not a 68% probability of a global crash.

A multi-asset fund: oil futures + S&P 500

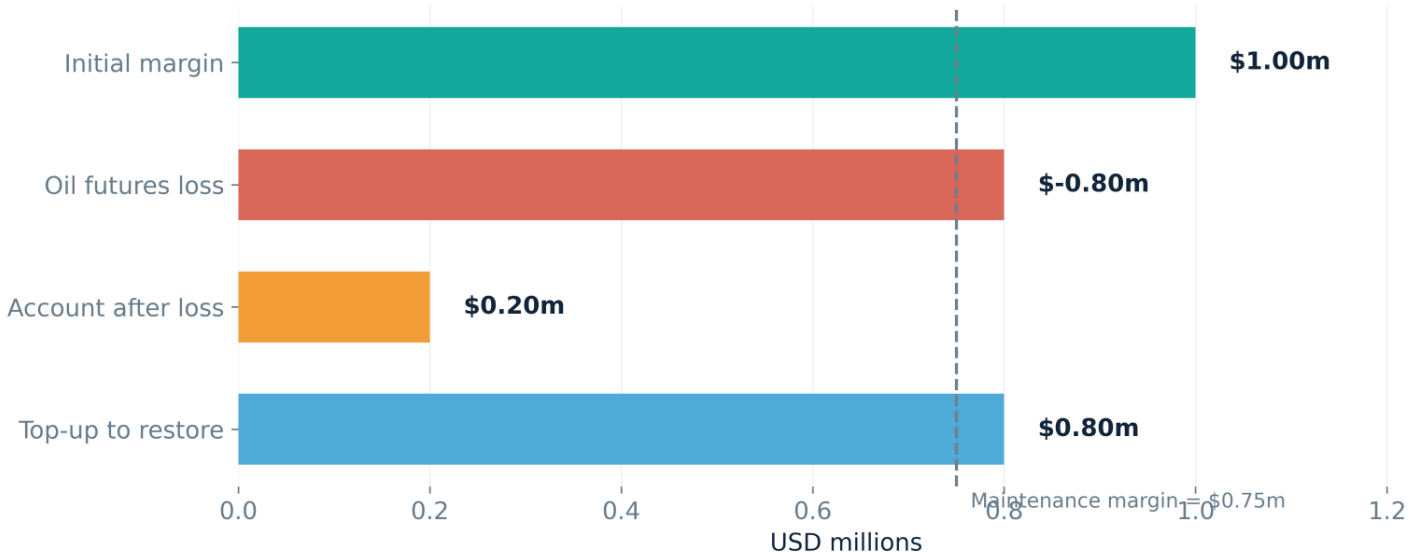
All amounts and dependence parameters on this slide are illustrative scenario assumptions.

CASE INPUT	VALUE	FINANCIAL MEANING
Oil futures notional	**$10.0m**	Long; initial margin $1.0m
S&P 500 exposure	**$10.0m**	Long equity investment
Calm Pearson correlation	**0.05**	Synthetic calm observations
Stress-tail dependence λL	**0.68**	Assumed Clayton θ ≈ 1.80
Stress one-day returns	**−8% / −7%**	Oil futures / equities

Do not treat these assumptions as verified historical market estimates.

Co-crash: $1.50m total loss and an $0.80m cash margin requirement

P&L and funding needs are distinct metrics that must not be double counted.



Economic loss

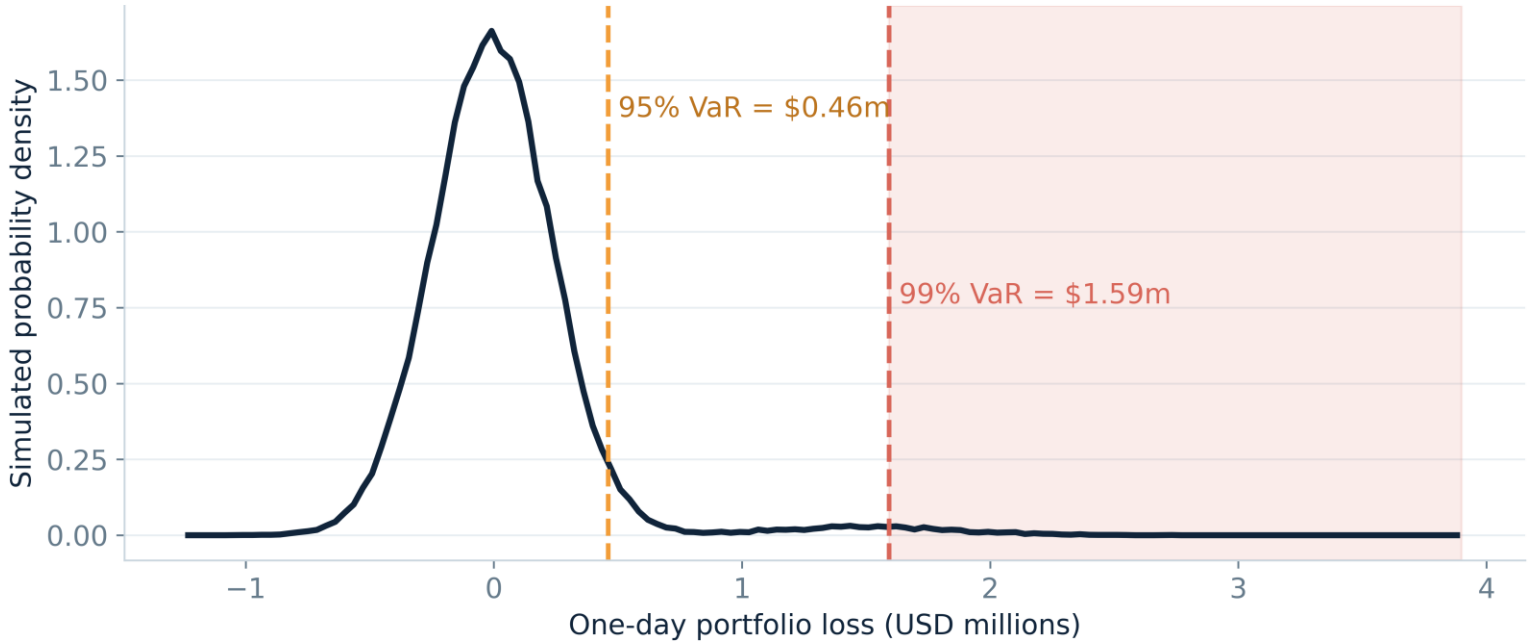
Oil futures: −$0.80m
Equity portfolio: −$0.70m
Total: −$1.50m

Liquidity response

Margin account: $1.00m → $0.20m. Replenishment: $0.80m. Keep reserves or reduce the futures exposure.

Model the distribution of one-day portfolio losses

Synthetic 80,000 outcomes: 97.5% calm and 2.5% stress regime; results are not backtested market forecasts.



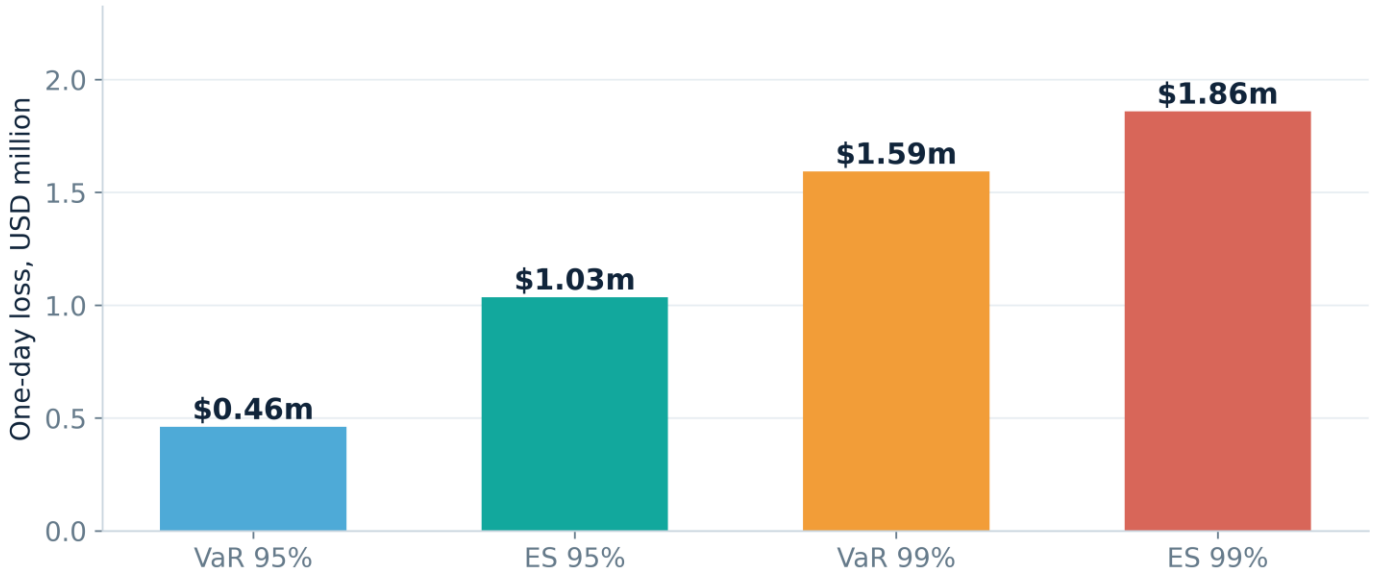
Portfolio loss formula

Negative of asset P&L. Exposure weights and loss sign convention must stay consistent.

$$L = -N_F\, r_F - N_E\, r_E$$

A 99% ES estimate must reflect tail severity

One-day, $10m oil futures + $10m equities; all figures USD millions, synthetic.



95% risk

VaR = $0.46m
ES = $1.03m

99% risk

VaR = $1.59m
ES = $1.86m

Use these as model outputs for discussion, not regulatory capital requirements.

From raw trading data to decisions

Dependence and tail modelling should not erase the need for cash stress testing.

01	**Validated data**	Futures settlements, rolls, equity returns
02	**Wavelet inspection**	Separate scales; retain relevant jumps
03	**EVT / POT fit**	Fit extremes and test threshold stability
04	**Copula simulation**	Generate joint asset-loss scenarios
05	**VaR, ES & cash**	Set limits and margin liquidity reserves

Sophistication does not eliminate model risk

A robust institution validates the assumptions before relying on the outputs.

DATA

Check contracts, maturities, roll returns and aligned dates.

WAVELETS

Test choice of basis, levels and retention of jumps.

EVT

Inspect threshold choice, sample size and ξ uncertainty.

COPULA

Compare Clayton against alternatives and regime changes.

BACKTEST

Test VaR exceptions and ES using suitable scoring methods.

LIQUIDITY

Stress margin calls, basis risk and forced liquidation.

The right decision is about losses AND liquidity

Five principles for finance learners and risk committees.

01 VaR is a threshold, not the maximum possible portfolio loss.

02 ES measures tail severity; EVT focuses on the far loss tail.

03 Wavelets reveal time scales, but economic meaning needs validation.

04 Copulas describe joint tail moves not captured by a single correlation.

05 **Futures margin creates liquidity risk: separate P&L from cash calls.**

Academic sources and important caveats

Use the references for theory; every plotted observation and case amount here is illustrative.

- Artzner, Delbaen, Eber & Heath (1999), “Coherent Measures of Risk,” Mathematical Finance, 9(3), 203–228.
- McNeil, Frey & Embrechts (2015), Quantitative Risk Management, 2nd ed., Princeton University Press.
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- Mallat (2009), A Wavelet Tour of Signal Processing, 3rd ed., Academic Press.
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- Hull (2022), Options, Futures, and Other Derivatives, 11th ed., Pearson.

Illustrative simulation only; not a claim about actual historical correlations, ξ, λ_L or regulatory capital.